

General HCF

from [unsatfit](https://unsatfit.github.io/unsatfit/hcmodel.html) document: <https://sekika.github.io/unsatfit/hcmodel.html>

The **general HCF** (hydraulic conductivity function) is defined as (Hoffmann-Riem et al, 1999)

$$K_r(h) = \frac{K(h)}{K_s} = S_e(h)^p \left[\frac{\int_0^{S_e(h)} h(S_e)^{-q} dS_e}{\int_0^1 h(S_e)^{-q} dS_e} \right]^r$$

where h is the pressure head (positive for unsaturated conditions), K is the unsaturated hydraulic conductivity, K_s is the saturated hydraulic conductivity, K_r is the relative hydraulic conductivity, and where p , q , and r are HCF parameters as explained below. S_e is effective saturation, defined by

$$S_e = \frac{\theta - \theta_r}{\theta_s - \theta_r}$$

, where θ is the volumetric water content, and θ_r and θ_s are the residual and saturated water contents, respectively.

As the HCF includes integral of the function $h(S_e)$, it is convenient when a closed-form expression of the integrated function is obtained for a specified **WRF** (water retention function) $\theta(h)$ or $S_e(h)$, as shown in [this page](#); otherwise numerical integration or approximation is required. Most of the **HCF in unsatfit** are closed-form expression of general HCF derived from respective WRF, as derived in Seki et al. (2022), and shown to be useful for practical applications in Seki et al. (2023).

1. HCF parameters

The general HCF expresses different type of models with HCF parameters, p , q , r as follows.

- **Burdine** (1953) model for $p=2$, $q=2$, $r=1$.
- **Mualem** (1976) model for $p=0.5$, $q=1$, $r=2$.

where Mualem's model is currently most widely used model. When p is used a variable and changed from the original value in those models, p is called a tortuosity factor.

The HCF depends on WRF. van Genuchten - Mualem model is the most popular selection of WRF and HCF models. However, it does not necessarily represent water retention and hydraulic conductivity curves in a wide range of pressure head. For giving more flexibility to the water retention curve, linear superposition model was proposed by Durner (1994) and Seki et al. (2022). Seki et al. (2023) showed that two parameters (p , q) or (p , r) should be optimized to represent hydraulic conductivity curves over a wide range of pressure heads.

In practice, as measuring (h , K) is difficult, HCF parameters are optimized to represent measured θ and/or h in water flow experiment by using such software as HYDRUS and PEST. See [Using unsatfit with HYDRUS](#) and [Using unsatfit with PEST](#) for detail.

2. Research history

- Burdine (1953) proposed a permeability model from pore-size distribution of porous system by applying [Hagen - Poiseuille equation](#).
- Brooks and Corey (1964) developed a WRF

$$S_e = \begin{cases} (h/h_b)^{-\lambda} & (h > h_b) \\ 1 & (h \leq h_b) \end{cases}$$

and HCF based on Burdine's model.

- Mualem (1976) proposed his model by considering the effect of combination of cylindrical tubes in radius r and ρ ; where the equivalent tube of radius R is expressed as $R^2 = r \rho$. Mualem compared his model with other models, including Burdine's model, with measured data of 45 soils. The value $p=0.5$ was obtained as the optimized value with these 45 soils. I note (Seki, 2022) that for Brooks and Corey model, Mualem's model ($p=0.5$, $q=1$, $r=2$) is equivalent to Burdine's model with changed p value ($p=1.5$, $q=2$, $r=1$).

- van Genuchten (1980) developed a WRF

$$S_e = [1 + (\alpha h)^n]^{-m}$$

and provided closed-form equations for Mualem's and Burdine's HCF. van Genuchten's WRF includes q as a parameter which is common with the HCF, and q determines the m - n relationship as $m = 1 - q/n$. Therefore, when Mualem's model is used, $q=1$ for WRF, and when Burdine's model is used, $q=2$ for WRF. As van Genuchten and Nielsen (1985) showed that Mualem's approach was found to be applicable to a wider variety of soils than Burdine's model, **van Genuchten-Mualem model** is the most widely used combination of WRF and HCF today, and in many papers it is written that $m = 1 - 1/n$ as $q=1$ is assumed.

- Hoffmann-Riem et al (1999) showed the general HCF model with van Genuchten WRF, and recommended optimizing p and r simultaneously.
- Kosugi (1996) proposed a WRF

$$S_e = Q \left[\frac{\ln(h/h_m)}{\sigma} \right]$$

where $Q(x) = \mathrm{erfc}(x/\sqrt{2})/2$ and derived Mualem's and Burdine's HCF, and showed that similar curves to van Genuchten model are obtained. Kosugi (1999) applied general HCF to his WRF and showed that p and q are better to be optimized simultaneously.

- Durner (1994) proposed a linear superposition of van Genuchten WRF, and showed that HCF calculated with Mualem's equation is better than simple van Genuchten model. Although Durner did not get a closed-form expression for HCF, Priesack and Durner (2006) solved the equation with general HCF. Durner's model with Mualem's model has been used in several studies for water flow simulation. Seki et al. (2022) generalized the calculation to other combinations of BC, VG and KO models.

Seki et al. (2023) showed that the equations represent hydraulic conductivity of broad range of soil types over a wide range of pressure heads, when two parameters (p , q) or (p , r) are optimized.